NITISH PATIL

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Summary

Final-year B.Tech CSE (AI/ML) student with strong DSA and OOP fundamentals, building backend systems and RAG/ML projects in Python, Java, and FastAPI, with an IEEE-published deep learning paper. Seeking a Software Engineer role.

Education

RV University, Bengaluru — B.Tech in CSE (AI & ML), CGPA: 8.64/10.0 2023 – 2027 (Expected)
Deeksha CFL PU College, Bengaluru — Pre-University (12th), PUE Karnataka: 90.66% 2023
Rashtrotthana Vidya Kendra, Dharwad — 10th, CBSE: 91.40% 2021

Publication

Multi-Disease Chest X-ray Classification and Explainability using Grad-CAM 24 Apr 2026
Research Paper, ICKECS 2026 (IEEE), 4 authors — Project: MedXAI | TensorFlow, Keras, CNN, Grad-CAM

- Conducted a comparative study of **15 CNN architectures** for four-class chest X-ray classification covering COVID-19, Pneumonia, Tuberculosis, and Normal cases.
- Investigated transfer learning and fine-tuning strategies; **InceptionResNetV2** achieved **93.51% accuracy**, improving to ∼**94%** after fine-tuning.
- Evaluated models with precision, recall, F1-score, and confusion matrices, using preprocessing, data augmentation, and class-imbalance handling to improve generalization.
- Applied **Grad-CAM** to generate visual explanations highlighting disease-relevant lung regions, improving the interpretability of model predictions.

Projects

VerifAI — Multimodal RAG Fact-Verification System (Team of 4) GitHub
LangChain, FastAPI, ChromaDB, BERT, Groq (Llama), React

- Built a fact-checking system that takes a claim, retrieves evidence from a 1,687-article news index, and returns a verdict with cited sources; falls back to Tavily web search and abstains only when results fail relevance and credibility checks — with the fallback off, it abstained correctly on **20/20** unanswerable claims and gave **zero** confidently-wrong verdicts across 200 FEVER claims.
- Fixed a token-truncation bug in the index and added hybrid dense + BM25 retrieval with a cross-encoder reranker, raising **recall@1 from 0.47 to 0.98** on the affected text.
- Fine-tuned two BERT classifiers on FEVER for claim and stance detection, used as pre-LLM filters so non-claims are rejected without an API call.
- Split the system into offline ingestion and online serving over a versioned index, with a manifest check that blocks startup on an embedding-model mismatch.

Dispatch Engine — Real-Time Delivery Dispatch System
Python, FastAPI, PostgreSQL, Redis, ARQ, WebSockets, SQLAlchemy, SSE

- Built a delivery dispatch backend that finds nearby available drivers, sends timed offers over WebSockets, and assigns an order to the first valid acceptance.
- Protected assignments against race conditions using PostgreSQL row locks and unique indexes, Redis locks, and idempotent requests; concurrency tests cover **50 drivers accepting one order** and **one driver accepting 50 orders**.
- Added a transactional outbox, Redis Streams, and ARQ jobs to publish events, handle offer timeouts, and recover pending work after interruptions.
- Implemented customer tracking through a REST status endpoint and live Server-Sent Events, including driver location updates and a straight-line ETA.

Technical Skills

Languages & Fundamentals: Java, Python, JavaScript; Data Structures, Algorithms, OOP
Frameworks & Libraries: FastAPI, React, LangChain, SQLAlchemy, TensorFlow, Keras
AI/ML: RAG, NLP, Transformers (BERT), CNNs, Grad-CAM
Databases: PostgreSQL, Redis, ChromaDB
Tools & Technologies: Git, GitHub, Postman, REST APIs, WebSockets, Server-Sent Events (SSE)

Extracurricular Activities & Achievements

- **Trek Lead, Art of Travel Club, RV University (2023 – 2025):** Led trekking expeditions of around 100 participants each, handling planning, logistics, safety, and cross-team coordination.
- **Achievements:** Smart India Hackathon — National Qualifier (2025); Deep Fake ML Challenge, IIIT Bengaluru (2025).